

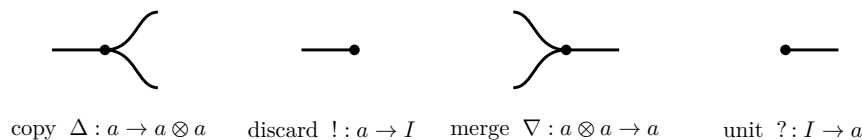
# §2.124 in string diagrams — a proof from the axioms

Freyd & Scedrov, *Categories, Allegories* §2.124:  $\text{Dom}(R \cap S) = 1 \cap SR^\circ$  — “a lemma we will use repeatedly”

An earlier note only **drew** the two sides and declared them equal. Here is the actual proof: the generators, the axioms they obey, two lemmas, and a step-by-step calculation in point-free (AOP) style, then drawn. Every algebraic step is tagged with the axiom it uses. The three underlying identities are proved in Lean for *arbitrary* relations — not a random sample — in `diag/S2_124.lean` (sorry-free, axioms `[propext]`).

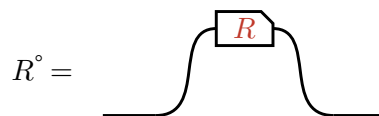
## §1. What a string diagram is

An arrow of a symmetric monoidal category drawn as a circuit: white boxes are the **relations**  $R, S : a \rightarrow b$ ; wires carry objects. Left-to-right juxtaposition is composition  $;$  (the book’s  $\times$   $y$ ), vertical stacking is the monoidal product  $\otimes$  (the book’s  $\oplus$ ). The identity  $1$  is a bare wire. An **allegory**’s operations are recovered from **one** extra structure on every object: a black-dotted **special Frobenius (co)monoid** — four generators.

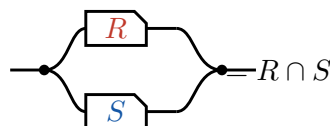


In REL these are the diagonal relations: **copy** is  $x \mapsto (x, x)$ , **discard** is  $x \mapsto \checkmark$ , **merge** =  $\text{copy}^\circ$ , **unit** =  $\text{discard}^\circ$ . Two derived pieces are used constantly — **converse** and **meet**:

**Converse**  $R^\circ$  is **free**: bend both of  $R$ ’s wires around, so what was an input is read as an output. In REL,  $x R^\circ y$  holds exactly when  $y R x$ .



**Meet**  $R \cap S$  is **not** a new generator; it is the convolution  $R \cap S = \Delta; (R \otimes S); \nabla$ : copy the input, run  $R$  and  $S$  in parallel, then **merge**. The merge forces the two outputs to **coincide** — it demands both  $xRy$  and  $xSy$ .



The **domain** used below is  $\text{Dom } R = 1 \cap R R^\circ$  (Freyd §2.122): the coreflexive  $\sqsubseteq 1$  picking out the  $a$  that  $R$  relates to **something**.

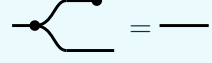
## §2. The axioms of the calculus

**Definition 2.1** (Special commutative Frobenius algebra (per object)).

The four dots obey these equations — the only moves a diagram may be rewritten by.  $\sigma$  is

the wire swap; **? is the unit** : a wire **created from nothing** ( $? : \mathbf{I} \rightarrow \mathbf{a}$ ), the converse of discard  $!$ . The pair  $(\Delta, !)$  copies and deletes; its converse pair  $(\nabla, ?)$  merges and creates. Each law is drawn beside it (the monoid  $(\nabla, ?)$  laws — assoc, unit, commutative — are the vertical mirrors of the first three):

**counit**  $\Delta; (1 \otimes !) = 1$

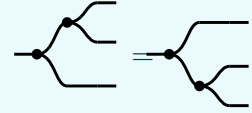


**cocommutative**  $\Delta; \sigma = \Delta$



**coassociative**  $\Delta; (\Delta \otimes 1) = \Delta; (1 \otimes \Delta)$

the picture re-brackets three wires for free; `delta_assoc` in the Lean spells the associator `tensAssoc` out

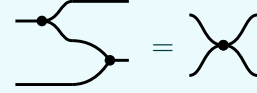


**special**  $\Delta; \nabla = 1$  *copy-then-merge = wire*



**Frobenius**  $(\Delta \otimes 1); (1 \otimes \nabla) = \nabla; \Delta = (1 \otimes \Delta); (\nabla \otimes 1)$

same suppression: `frob_left` / `frob_right` carry `tensAssoc` too, because the tensor is not strictly associative



**Converse**  $(-)^{\circ}$  is the involution reversing composition and fixing the structure:

$$(P; Q)^{\circ} = Q^{\circ}; P^{\circ}, \quad P^{\circ\circ} = P, \quad \Delta^{\circ} = \nabla, \quad !^{\circ} = ?, \quad (P \otimes Q)^{\circ} = P^{\circ} \otimes Q^{\circ},$$

whence  $(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$ . With `cup` =  $?; \Delta$  and `cap` =  $\nabla; !$ , the special and Frobenius laws give the **yanking** (snake) equations, so a wire may be straightened.

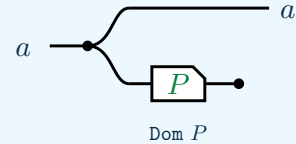
**Derived — spider fusion.** From the four groups above, any **connected** diagram built only from  $\Delta, \nabla, !, ?$  with  $m$  inputs and  $n$  outputs equals the **single spider**  $s_{(m,n)}$ . This is the one power tool: a tangle of dots collapses to one node with the right number of legs.

### §3. Two lemmas

**Lemma 3.1** (domain = copy · run · discard).

$$\text{Dom } P = 1 \cap PP^{\circ} = \Delta; (1 \otimes P); (1 \otimes !).$$

Copy the input; run  $P$  on one copy; **discard** its output. The discard is the existential “ $P$  relates to something”, so this is the domain coreflexive. (Both sides denote  $\{(a, a) : \exists b. aPb\}$ .)

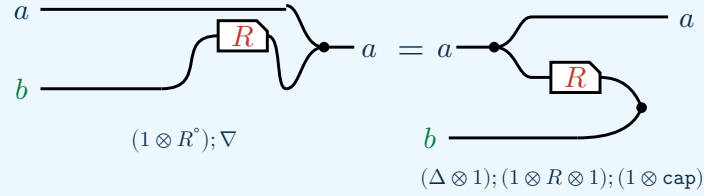


**Lemma 3.2** (converse folds into a cap).

 For  $R : a \rightarrow b$ ,

$$(1 \otimes R^\circ); \nabla = (\Delta \otimes 1); (1 \otimes R \otimes 1); (1 \otimes \text{cap}).$$

A converse  $R^\circ$  feeding a merge equals: **copy** the surviving wire, run  $R$  forward, and **cap** its output against the incoming one — just “bend  $R$ , then yank” (converse + Frobenius). This is the move that turns  $S R^\circ$  into a witness.



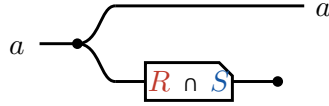
## §4. The proof

**Theorem 4.1.**

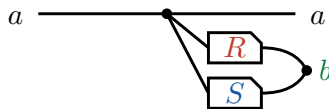
$$1 \cap SR^\circ = \text{Dom}(R \cap S).$$

Read the pictures top to bottom:  $\text{Dom}(R \cap S)$  and  $1 \cap S R^\circ$  both reduce to the **same** middle diagram  $W$ , so  $\text{Dom}(R \cap S) = W = 1 \cap S R^\circ$ . Each  $=$  is one graphical move.

**Dom(RnS)** Lemma 1: copy  $a$ , run  $R \cap S$  on one copy, discard

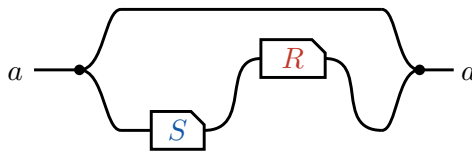


$=$  unfold  $R \cap S = \Delta; (R \otimes S); \nabla$ ;  $\nabla; ! = \text{cap}$ ; fuse the copies (coassoc)



the normal form  $W = \Delta^3; (1 \otimes R \otimes S); (1 \otimes \text{cap})$  tap  $a$ ; run  $R, S$  cap to one  $b$

$=$  Lemma 2: bend the  $R^\circ$  straight



**1 ∩ S R°** copy; the top wire is the  $1$ ;  $S$  then the bent  $R^\circ$ ; merge

The one step with content is **Lemma 2** (bottom  $=$ ): bending the  $R^\circ$  in  $S R^\circ$  straightens the  $S$ – $R^\circ$  loop into  $R$  beside  $S$ , both fed by the same tap, their  $b$ -outputs meeting at the

cap — the shared witness  $b$ . Everything else is unfolding  $\eta$  and spider fusion. No modular law appears; §2.124 is here **derived** from Frobenius + converse.

## §5. Is this checked in Lean?

Yes — every boxed result above is a Lean theorem in `diag/S2_124.lean`, sorry-free, for arbitrary relations  $R, S$ :

| in this note                                        | in <code>diag/S2_124.lean</code>                                                                |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Lemma 3.1 (domain = copy · run · discard)           | <code>dom_cd</code>                                                                             |
| Lemma 3.2 (converse folds into a cap)               | <code>cv_merge</code>                                                                           |
| Theorem 4.1 $1 \cap SR^\circ = \text{Dom}(RnS)$     | <code>dom_inter_diag</code> — via <code>left_eq_W</code> , <code>right_eq_W</code>              |
| the axioms of §2 (special, Frobenius, coassoc, ...) | the <code>CartBicat</code> fields of <code>diag/CB.lean</code> , and <code>special</code> there |

How Lean checks them. Not in a concrete model: the statements are made over an arbitrary **cartesian bicategory of relations** (`diag/CB.lean`, Definition 4.1) and the proofs ARE the diagram rewriting above, step for step. Nothing is suppressed — the associators and unitors a picture hides are written out, which `diag/Monoidal.lean` supplies. `diag/RelSetCB.lean` then instantiates all of it at this repo's `Rel(Set)`.

The  $1nPP^\circ \rightarrow \Delta; (1\otimes P); !$  collapse of Lemma 1 was, in an earlier version of this file, an extra relational axiom called `adequacy` ( $\Delta; (R\otimes R); \nabla = R$ ). It is not an axiom: it is `meet_idem`, a theorem of Definition 4.1.

One genuine use of the modular law survives, inside Lemma 1 —  $1 \cap P\tau \sqsubseteq PP^\circ$ , which is `modular_right` at  $S := \tau$ . The modular law is itself a theorem here (`modular_of_frobenius`), derived from the Frobenius equation and the lax copy inequation, so the calculus is not assuming it.

The allegory-level §2.124, which Freyd proves from the modular law directly, is `dom_inter` in `Freyd/S2_10.lean`; its two halves are the two moves this diagram fuses into one:

| <code>dom_inter</code> step (allegory / AOP)                                                                | string-diagram move                                                                  |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $1 \cap SR^\circ \sqsubseteq 1 \cap (RnS)R^\circ$<br>— <code>modular_le S R^\circ 1</code>                  | Lemma 2: bend $R^\circ$ , so $R$ hangs beside $S$                                    |
| $= 1 \cap R(R^\circ nS^\circ)$ — a coreflexive equals its reciprocal                                        | read the tapped loop either way (converse of a coreflexive = itself)                 |
| $\sqsubseteq (RnS)(R^\circ nS^\circ) = \text{Dom}(RnS)$<br>— <code>modular_le R (R^\circ nS^\circ) 1</code> | coassoc: the two prongs share one tap $\Delta^3$ and one <b>cap</b>                  |
| the outer $1$ kept at each step ( <code>le_inter</code> with $\sqsubseteq 1$ )                              | the straight <b>diagonal wire</b> — the spider only <b>taps</b> it, never deletes it |